



Figure 5: Select Migration Method



Figure 6: Create MySQL target object



Figure 7: Migrate Data to MySQL



Figure 8: Review Migration

The answers to these questions will pretty much sum up your requirements. Now that you have the clarity of what you need to achieve, you can actually plan to fill the gap and achieve it, tangibly. Let's look at the process of migration—the migration life-cycle.

The migration life-cycle

The migration process primarily contains five distinct steps:

1. Document the source:

Document the source from where you want to migrate your database. How many objects does it contain? What is the relation between these objects? What are the different rules?

2. Define targets:

Define the targets for migration. How many objects do you want from your source? What relations do you want from the existing database? Do you want to modify any

existing relation? Or drop/add any existing relation/rules?

3. **Design the ETL flow:** Design the ETL flow for migration. How do you want to extract? How do you want to transform the objects?
4. **Migrate:** After successfully completing the above steps, execute the migration plan.
5. **Test performance:** After successful migration of your database, you must check the performance of the system. It should be at par with what was planned.

Remember to optimise your MySQL installation—many administrators make the common mistake of not optimising their MySQL installations, and that results in a performance drop after some time. Don't be beguiled into believing that it is MySQL that is not performing.

After all this theory, let's look at an example of how to migrate from Oracle to MySQL.

Data migration from Oracle

I'm not saying that Oracle is not a good RDBMS—it's great, but proprietary. This is just an example that will introduce you to the two possible ways—manual and automated—of migrating from one RDBMS to another.

In a manual migration, you manually take care of things and follow the migration life cycle step-by-step, backing things up, mapping and deploying, manually. In an automated migration, you use tools to help you.

So how does the manual system work? The process is as follows:

1. Get equivalent data types from Oracle to MySQL.
2. Get equivalent functions
3. Get equivalent syntaxes
4. Find out alternates, if no option